

Compositional monitoring of energy-mix drift on the simplex

Five viewpoints. One observable stack. The hidden drivers, named.

CoDaWork 2026 · 11th International Workshop on Compositional Data Analysis

Coimbra, Portugal · 1–5 June 2026

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Manuscript available — see Standard Stamp on final slide.

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.

The Question

What carriers really drove each country's energy transition — and which were doing the structural work?

When an electricity mix shifts over twenty-five years, two questions appear together:

1. WHAT path did the system take?

Which carriers grew, which shrank, which stayed flat?

2. WHY did it take that path?

Which carriers were doing the structural work behind the change?

*The standard answer is a stacked-area chart of share-over-time.
It answers the first question and silently misleads the second.*

A small-share carrier whose share triples appears as a sliver.
A large-share carrier drifting one point appears as the headline.

This talk introduces the Activation Coefficient — the formal name for the carrier-level diagnostic that names the hidden drivers the size view never shows.

Viewpoint 1 — the size view

What everyone sees when they look at the size view. And what they miss.

WORLD ELECTRICITY GENERATION — the standard stacked-area chart

- Coal stays dominant for most of the period.
- Solar appears as a thin yellow sliver after 2010.
- Gas grows substantially.
- Nuclear declines slowly.
- Wind grows steadily but is small until late.

What the chart hides:

USA, 2012–2013.

Solar holds 0.107 % of the mix.

Solar did 81.7 % of the structural work.

Activation Coefficient: $760 \times$ its size.

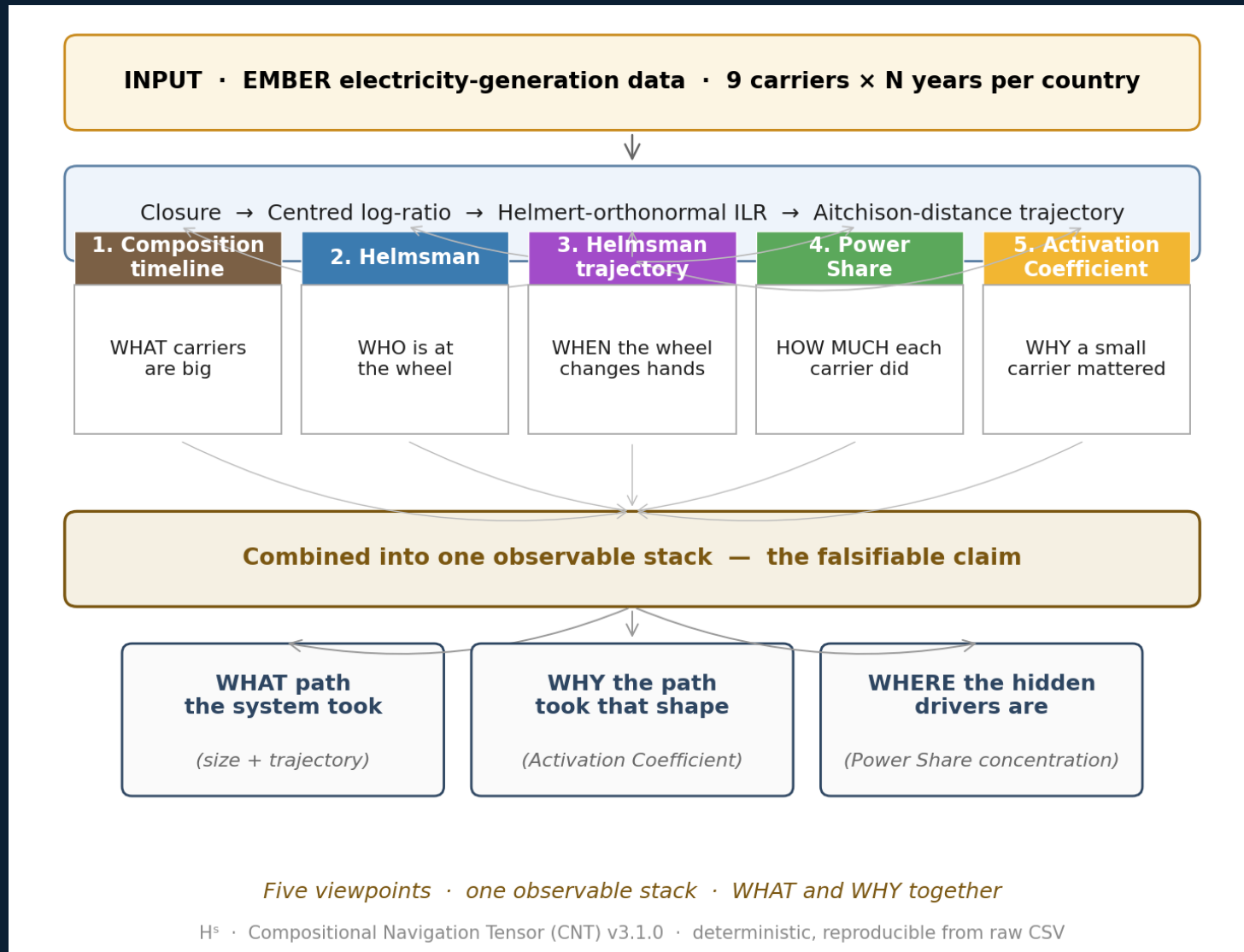
The size view did not show that. No size view ever could.

The protocol presented today reads compositions natively in Aitchison geometry, stacks five viewpoints into one observable, and names the hidden drivers explicitly.

Mathematical foundation: standard compositional data analysis. Operational claim: monitoring application is what is new.

The five-viewpoint protocol

Each viewpoint answers one specific question. Combined, they yield a complete answer for any year-to-year transition step.



Viewpoint 2 — the helmsman

Who is at the wheel each year? The single carrier with the largest CLR displacement at each transition step.

FORMAL DEFINITION (Eq. 5)

$$\sigma(t) = \operatorname{argmax}_i | \operatorname{clr}_i(t+1) - \operatorname{clr}_i(t) |$$

The helmsman is a categorical assignment per step. It names which carrier did the most directional work between two consecutive years.

READING THE HELMSMAN

- The helmsman trajectory $\sigma(1), \sigma(2), \dots$ shows who held the wheel year-by-year.
- The 'flip count' is the number of times the helmsman changes carrier across the window.
- Plotted with DOTTED lines — no continuous carrier-to-carrier path is implied.
- The helmsman is a categorical label, not a position on a continuum.

In our corpus the helmsman flip count ranges from 4 (World aggregate, stable) to 17 (Japan, post-Fukushima cascade).

Viewpoint 3 — Power Share

How much each carrier did at each transition. Sums to 100 % across the carriers at every step.

FORMAL DEFINITION (Eq. 6)

$$\pi_i(t) = (\Delta \text{clr}_i)^2 / \sum_j (\Delta \text{clr}_j)^2, \quad \sum_i \pi_i(t) = 1$$

The natural decomposition of the squared Aitchison distance across the carriers. No carrier is hidden by the bookkeeping.

WHY IT MATTERS

- The standard size view tells you what's BIG.
- The Power Share tells you what's MOVING.
- These are different questions. A 30 %-share carrier holding steady has 0 % Power Share.
- A 0.1 %-share carrier in rapid growth can have 70–85 % Power Share at the step.

What's big / what's moving. This is the operational distinction that makes the rest of the protocol useful.

Viewpoint 4 — the Activation Coefficient (the yeast factor)

Why a small carrier mattered. The leverage-to-size ratio — directional work divided by compositional weight.

FORMAL DEFINITION (Eq. 7)

$$\alpha_i(t) = \Pi_i(t) / \rho_i(t) \quad \text{reported when } \rho_i \geq 0.1 \%$$

Power Share divided by the carrier's composition share at the start of the step.

INTERPRETATION

- $\alpha = 1$ the carrier did exactly its size's share of work
- $\alpha \gg 1$ a small carrier doing structural work far beyond its size → hidden driver → yeast
- $\alpha \ll 1$ a large carrier moving less than its size would predict → structural ballast

WORKED EXAMPLE — USA Solar 2012 → 2013

composition share start of 2012	:	0.107 %	(solar was a tiny sliver)
power share of squared CLR motion	:	81.7 %	(solar did most of the work)
activation coefficient	:	760 ×	(solar punched at 760× its size)

The hidden driver — solar, 2010–2015

Across nine national electricity mixes, solar at sub-0.2 % composition share did 70–85 % of the structural directional work.

TOP YEAST MOMENTS • 9-COUNTRY CORPUS • filtered to composition share ≥ 0.1 %

#	Country	Transition	Carrier	AC × size	Power Share	Size at start
1	USA	2012–2013	Solar	760 ×	81.7 %	0.107 %
2	FRA	2010–2011	Solar	659 ×	72.6 %	0.110 %
3	FRA	2004–2005	Wind	634 ×	66.0 %	0.104 %
4	CHN	2013–2014	Solar	549 ×	84.7 %	0.154 %
5	WLD	2010–2011	Solar	513 ×	77.7 %	0.151 %
6	FRA	2005–2006	Wind	495 ×	83.5 %	0.169 %
7	USA	2013–2014	Solar	395 ×	87.9 %	0.223 %
8	CHN	2004–2005	Bioenergy	346 ×	39.7 %	0.115 %
9	DEU	2005–2006	Solar	333 ×	71.1 %	0.214 %
10	GBR	2001–2002	Wind	190 ×	47.5 %	0.249 %

Solar dominates seven of the top ten. All in the 2010–2015 window. Composition share between 0.1 % and 0.2 %.

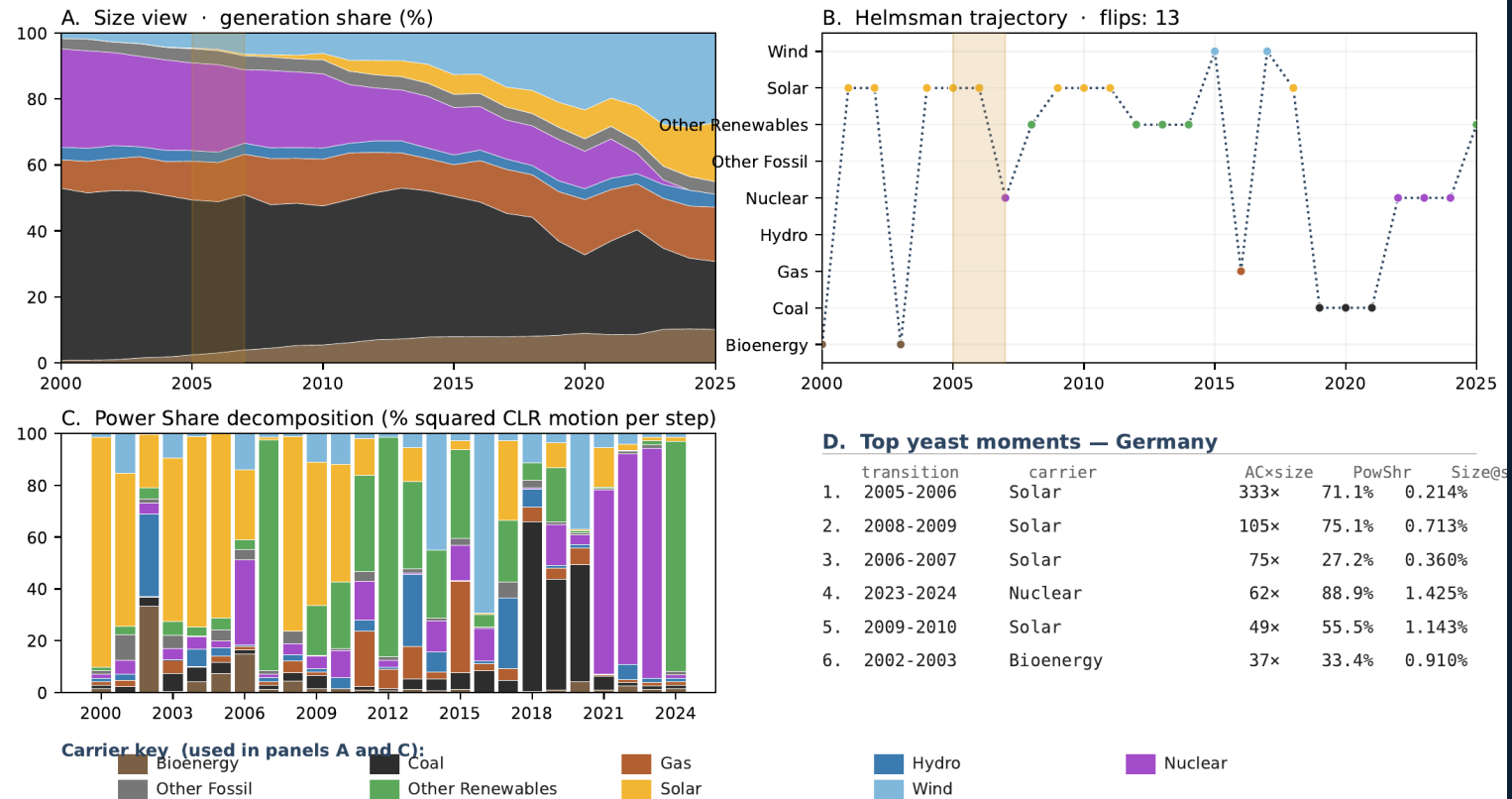
The size view never showed it. The Activation Coefficient names it.

Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The acceleration moments hidden in the size view.

Germany — the continuous arc to the renewable vertex

Twenty-five years of deliberate composition change. The size view shift is real — the yeast moments show when it actually accelerated.



Germany's transition is the textbook case of a continuous arc: coal recedes, gas holds, solar and wind grow steadily.

But the Power Share view shows the acceleration moments — solar in 2005-2006 at 0.21 % composition share doing 71 % of the structural directional work (activation coefficient $\approx 333\times$).

That is the structural beginning of the Energiewende — three years before the size view would call it visible.

Hidden Drivers in Energy Mixes · H⁺ CNT v3.1.0 · Aitchison-geometric protocol

Fig. 2

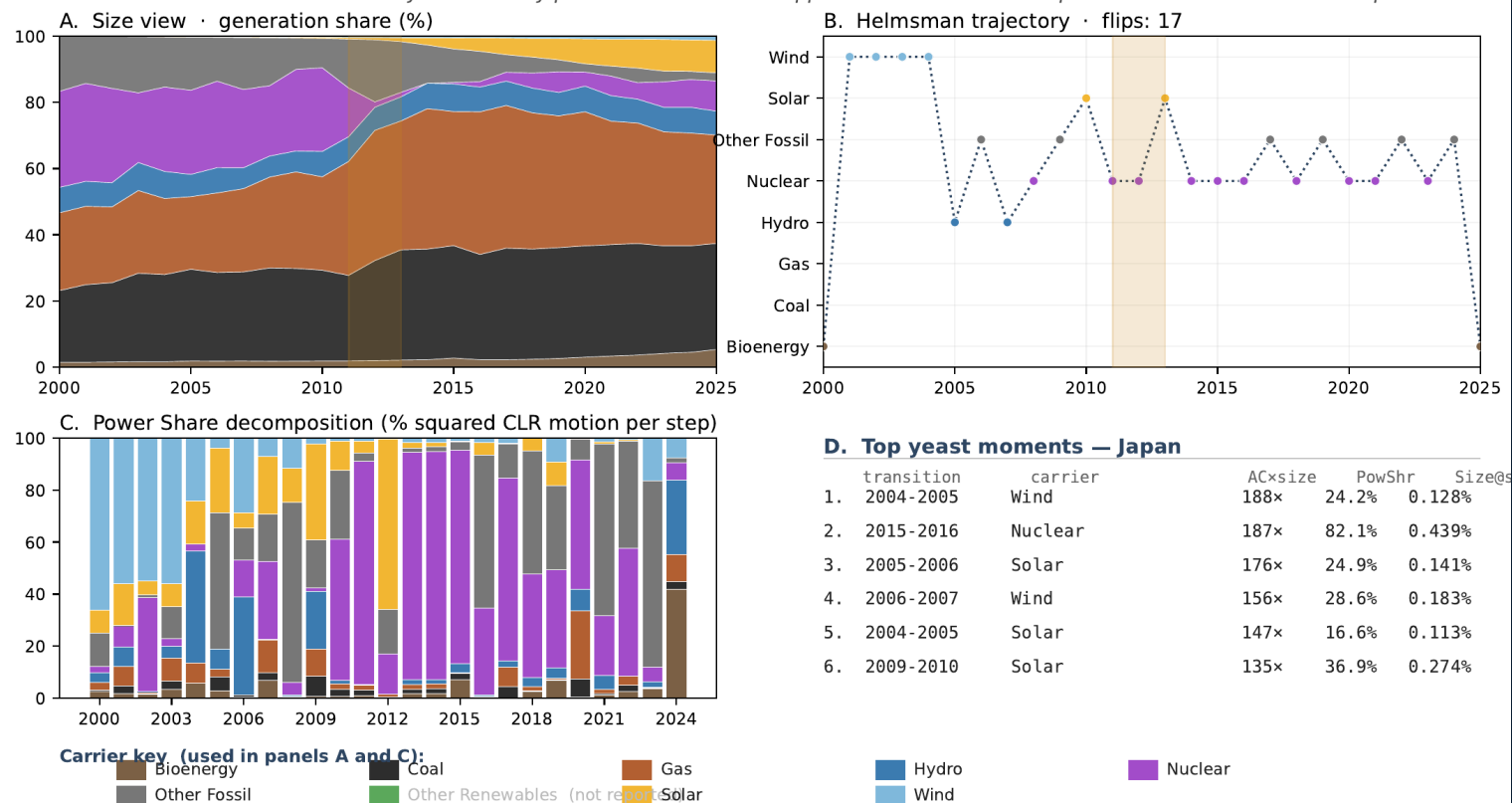
Solar 2005-2006 at 0.21 % share, doing 71.1 % of the work — $AC \approx 333\times$. The structural beginning of the Energiewende.

Japan — Fukushima 2011 and the multi-year reorganisation

An external shock that registers in every viewpoint. The post-shock cascade tells the deeper story.

Japan — Fukushima 2011 and the structural aftermath

An external shock and the multi-year recovery path. The helmsman flipped 17 times across the period — the loudest in the corpus.



The 2011-2012 transition shows in every viewpoint at once: composition (nuclear collapses), helmsman (carriers shift abruptly), Power Share (massive concentration of squared CLR motion at one step).

The years AFTER tell the deeper story — Japan's post-Fukushima mix kept reorganising for over a decade, and the yeast moments distribute across multiple carriers as solar, gas, and renewables each took turns absorbing the displaced nuclear share.

Hidden Drivers in Energy Mixes · H* CNT v3.1.0 · Aitchison-geometric protocol

Fig. 3

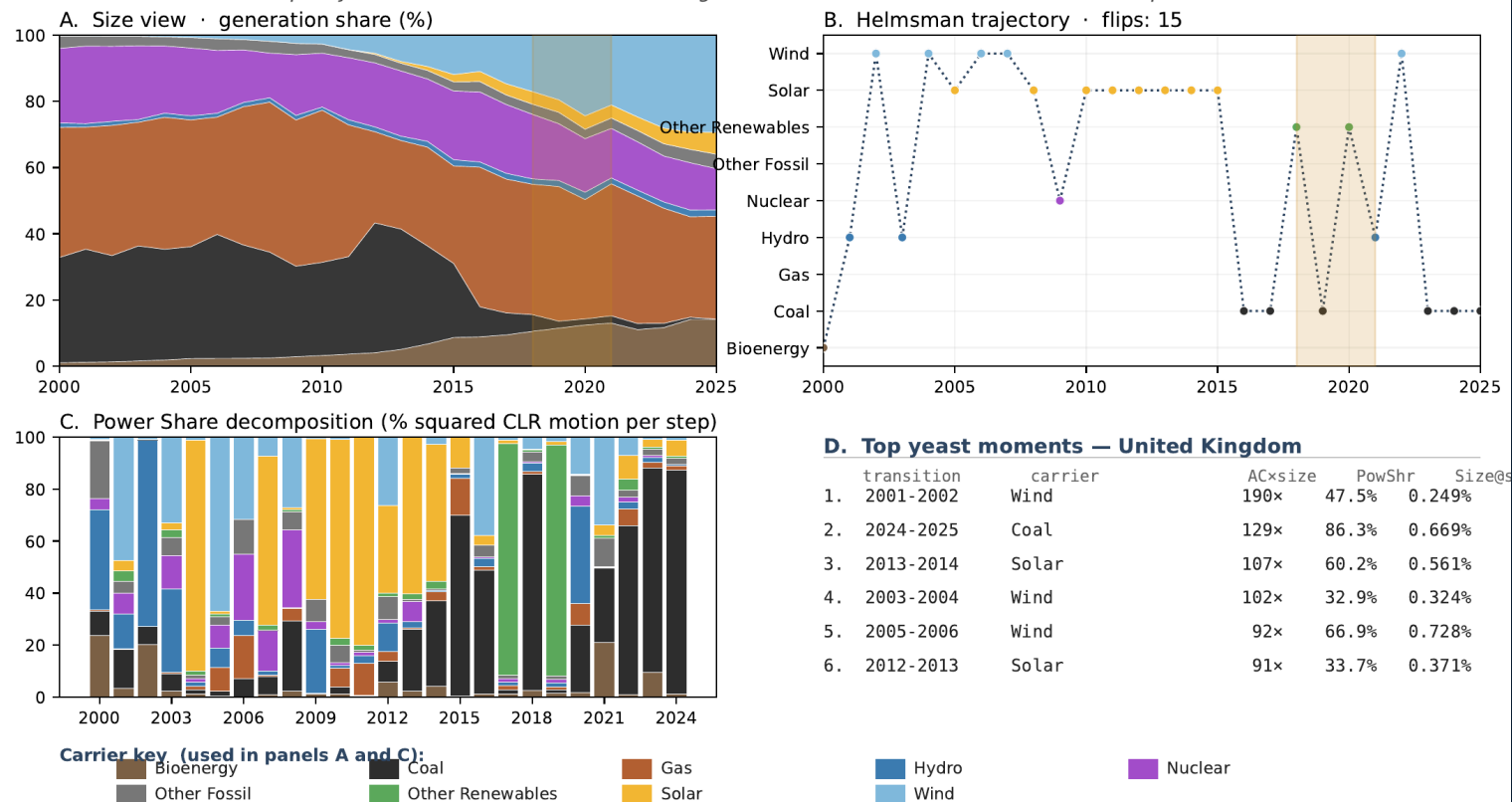
Helmsman flips: 17 — the loudest in the corpus. Aitchison distance 2011 → 2012 = 3× neighbouring years. Multi-year cascade.

United Kingdom — the coal exit as a regime change

Policy-driven transition. Coal goes from > 30 % to < 2 %. Specific small carriers absorbed the displaced share.

United Kingdom — the coal exit as a regime change

A policy-driven structural transition. Coal goes from dominant to near-zero across the period.



The UK story is the abrupt-policy archetype. Coal's share drops from over 30 % to under 2 % across the period.

The Power Share view names what made it possible: Other Renewables and Wind in the late-2010s took turns doing the structural work of absorbing coal's exit, often at low single-digit composition share (panel D).

Without the activation coefficient, those small-share carriers are invisible. With it, the whole transition snaps into focus.

Hidden Drivers in Energy Mixes · H⁺ CNT v3.1.0 · Aitchison-geometric protocol

Fig. 4

Coal disappeared into specific, small, hard-working sub-categories — not into 'renewables' broadly.

Germany — the navigation chart

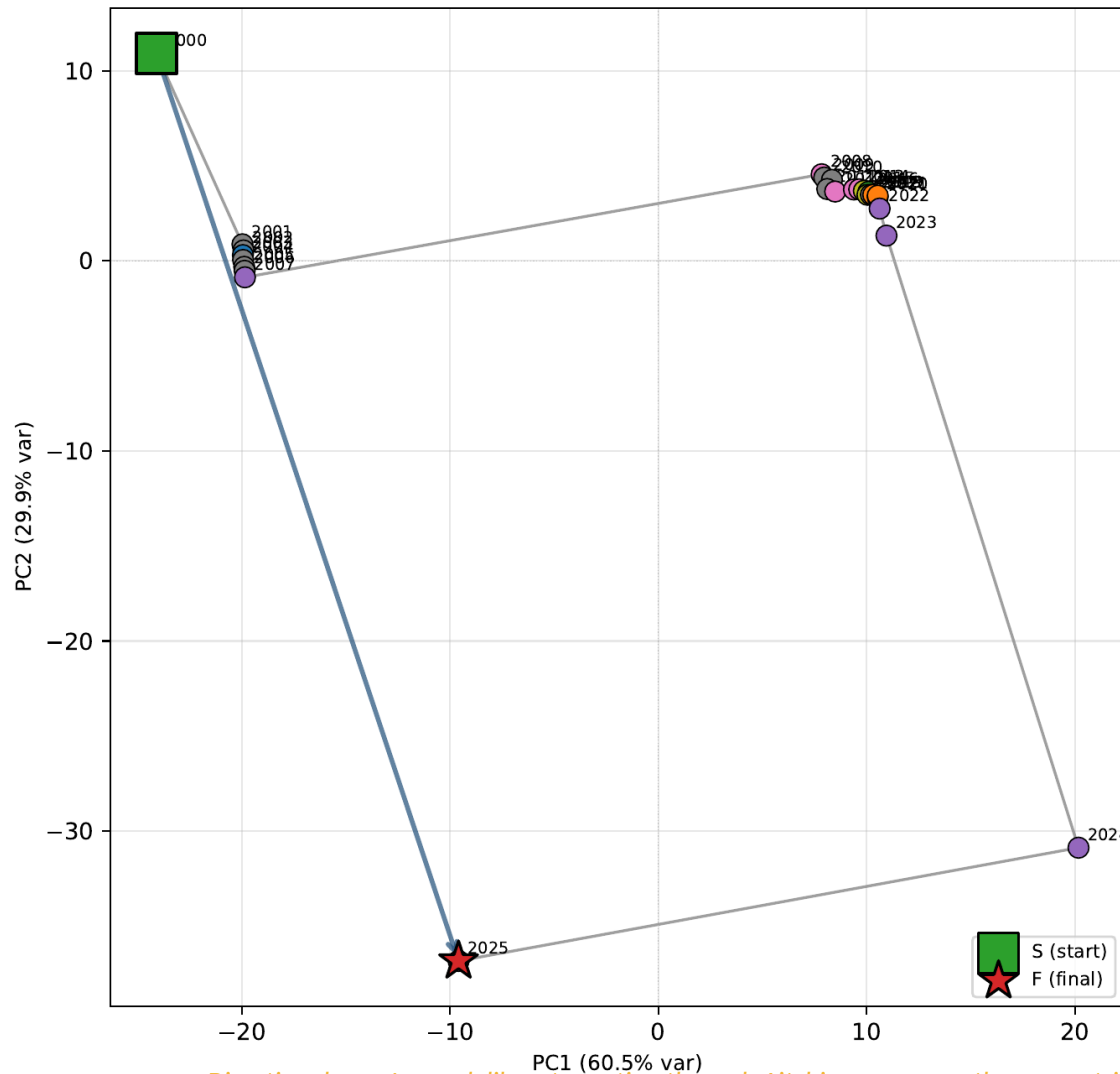
PCA 2D projection of the CLR trajectory. Continuous-arc archetype — course directness 0.41.

Compositional system geometry and dynamics for, Germany — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2

CLR trajectory projected by PCA



Navigation metrics ($V_{\text{net}} = h_F - h_S$)

h_start label : 2000

h_final label : 2025

net_distance : 55.7717 HLR

path_length : 136.5882 HLR

course_directness: 0.4083

(1.0 = straight line, 0 = pure looping)

dynamic range S : 40.2314 HLR

dynamic range F : 39.4517 HLR

Reading guide:

tight cluster → stable composition

long excursion → structural shift

sharp turn → regime transition

loop → cyclic behaviour

spiral → systematic drift

Directional arc. Long, deliberate motion through Aitchison space — the geometric signature of the Energiewende.

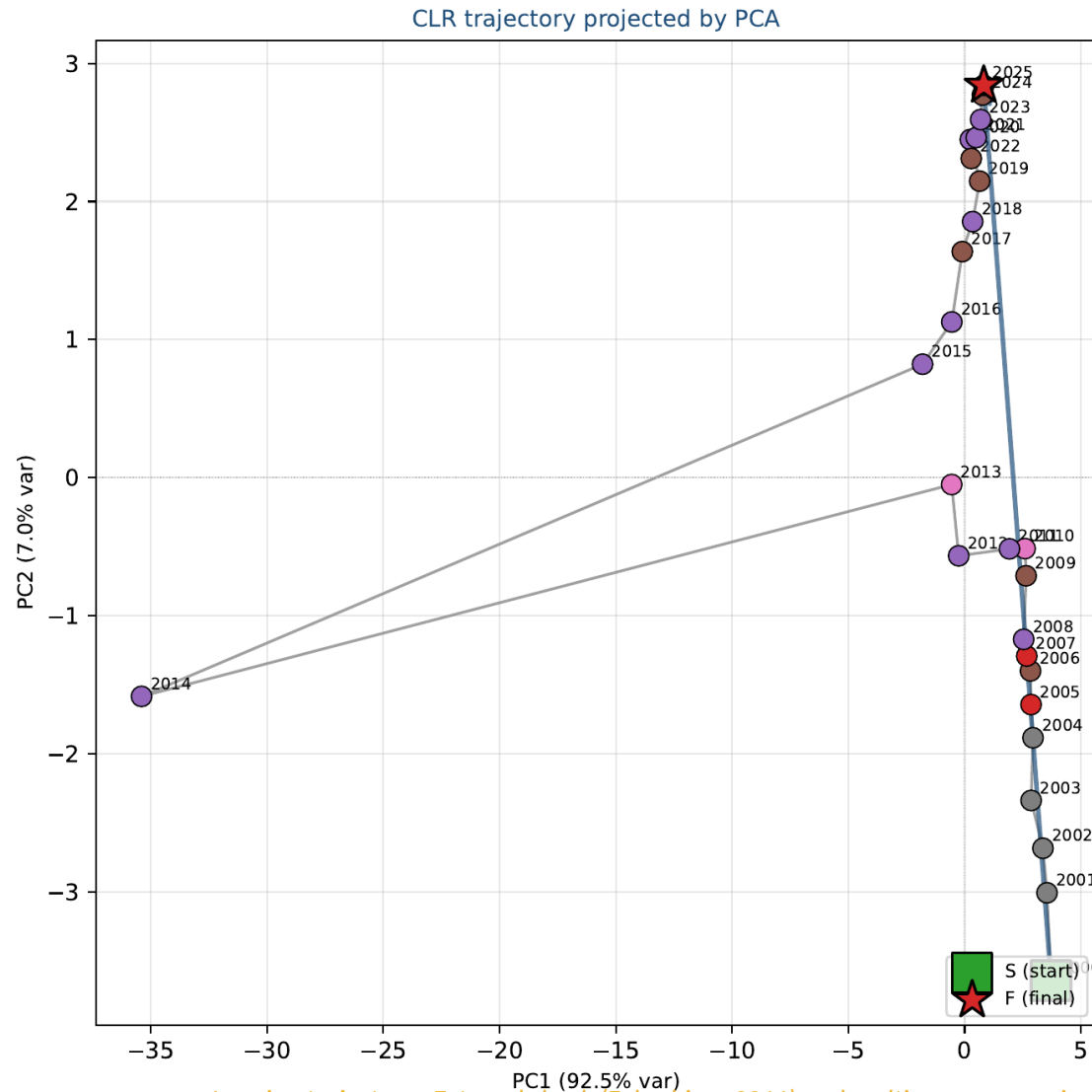
Japan — the navigation chart

PCA 2D projection of the CLR trajectory. Heavy-looping archetype — course directness 0.09.

Compositional system geometry and dynamics for, Japan — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Navigation metrics ($V_{net} = h_F - h_S$)

h_start label : 2000

h_final label : 2025

net_distance : 7.2233 HLR

path_length : 82.5468 HLR

course_directness: 0.0875

(1.0 = straight line, 0 = pure looping)

dynamic range S : 7.9728 HLR

dynamic range F : 3.2684 HLR

Reading guide:

tight cluster → stable composition

long excursion → structural shift

sharp turn → regime transition

loop → cyclic behaviour

spiral → systematic drift

Looping trajectory. External shock (Fukushima 2011) and multi-year reorganisation register as path curvature.

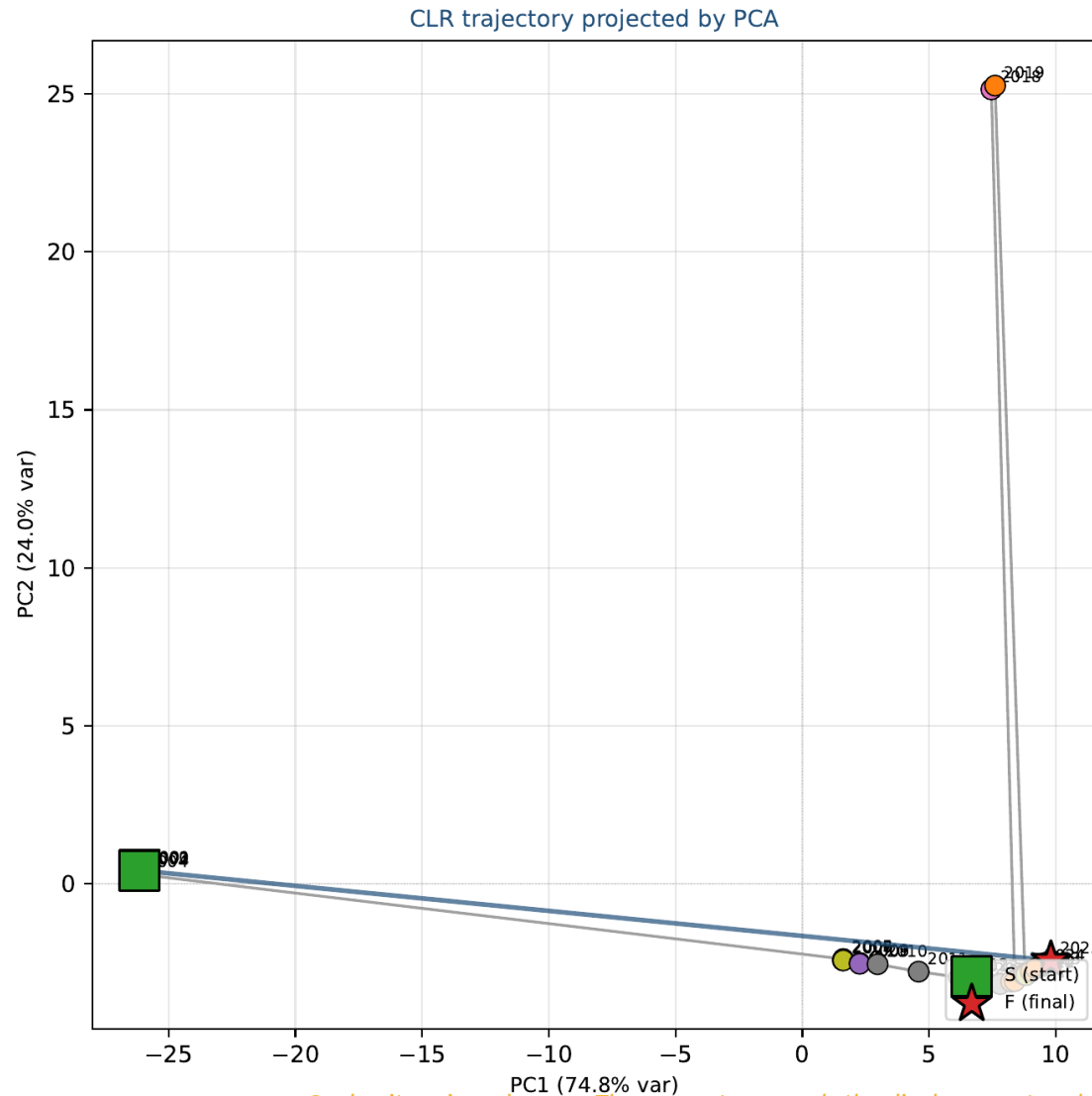
United Kingdom — the navigation chart

PCA 2D projection of the CLR trajectory. Jump-and-return archetype — course directness 0.36.

Compositional system geometry and dynamics for, United Kingdom — EMBER electricity generation 2000-2025

Plate 16 | System Course Plot — PCA 2D path with S→F net vector

Engine cnt 2.0.4 | Schema 2.1.0 | Order 2



Navigation metrics ($V_{\text{net}} = h_F - h_S$)

h_start label : 2000

h_final label : 2025

net_distance : 36.2591 HLR

path_length : 100.3591 HLR

course_directness: 0.3613

(1.0 = straight line, 0 = pure looping)

dynamic range S : 39.5365 HLR

dynamic range F : 39.0486 HLR

Reading guide:

tight cluster → stable composition

long excursion → structural shift

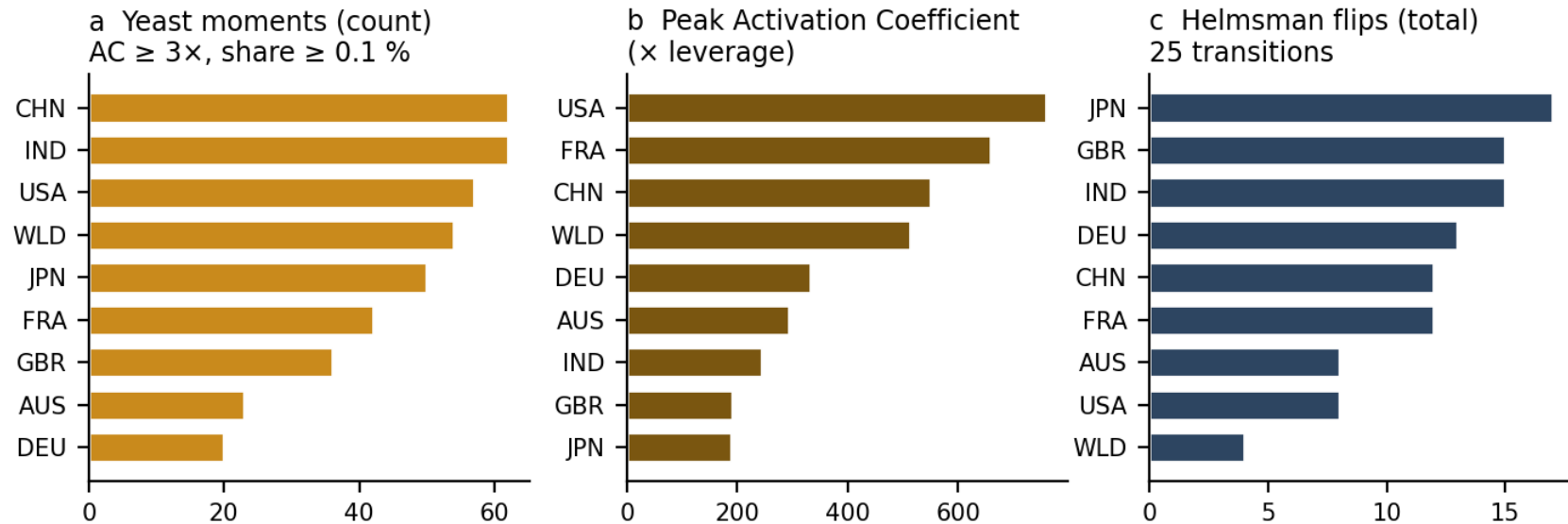
sharp turn → regime transition

loop → cyclic behaviour

spiral → systematic drift

Cross-country signature — five of nine countries reproduce the deceptive-drift pattern

Australia, China, United Kingdom, India, Japan fire the signature. Germany, France, USA, World aggregate do not.



*Japan tops helmsman flips (17 — the shock). China & India top yeast count (62 each — small carriers absorbed).
USA, France, World aggregate top peak Activation Coefficient — solar 2010-2015 at 500-760 $\times$.*

d Top ten activation moments across the 9-country corpus

#	country	transition	carrier	Activation	Power Share	Size at start
1.	USA	2012-2013	Solar	760 $\times$	81.7 %	0.107 %
2.	FRA	2010-2011	Solar	659 $\times$	72.6 %	0.110 %
3.	FRA	2004-2005	Wind	634 $\times$	66.0 %	0.104 %
4.	CHN	2013-2014	Solar	549 $\times$	84.7 %	0.154 %
5.	WLD	2010-2011	Solar	513 $\times$	77.7 %	0.151 %
6.	FRA	2005-2006	Wind	495 $\times$	83.5 %	0.169 %
7.	USA	2013-2014	Solar	395 $\times$	87.9 %	0.223 %
8.	CHN	2004-2005	Bioenergy	346 $\times$	39.7 %	0.115 %
9.	CHN	2003-2004	Bioenergy	342 $\times$	45.5 %	0.133 %
10.	FRA	2024-2025	Coal	342 $\times$	62.1 %	0.182 %

The synthesis — WHAT path + WHY

Five viewpoints stack into one observable. Each gives a partial glimpse; together they form the complete picture.

Viewpoint 1 • Size view

WHAT carriers are big

Standard stacked-area chart

Viewpoint 2 • Helmsman

WHO is at the wheel

Categorical per step

Viewpoint 3 • Helmsman trajectory

WHEN the wheel changes

Dotted-line plot over time

Viewpoint 4 • Power Share

HOW MUCH each carrier did

Squared-CLR decomposition (sums to 100 %)

Viewpoint 5 • Activation Coefficient

WHY a small carrier mattered

Power Share ÷ composition share

Used together, the five viewpoints answer WHAT • WHO • WHEN • HOW MUCH • WHY — the complete answer.

The falsifiable claim — MC-4 in three conjuncts

Aitchison-native + formal change detection + carrier-level attribution → one observable stack.

Four explicit ways a CoDa specialist could defeat the claim:

1. Prior-art defeat

Show the three-conjunct combination already exists in the literature. Closest adjacent prior art identified: Morais, Thomas-Agnan & Simioni (2018); Arata & Onozaki (2017). Neither combines all three to our reading; a closer match is welcome.

2. Metric defeat

Show the verdicts reverse under a different valid simplex metric. We have verified pair-invariance for TV distance and Aitchison distance across 101 datasets — but not exhausted the family.

3. Case defeat

Show the 5-of-9 deceptive-drift signature is an artefact of preprocessing, carrier definition, or null-model choice rather than a robust compositional read. Full corpus is public.

4. Category defeat

Show compositional monitoring is most accurately an application note inside existing CoDa rather than a distinct monitoring category. We hold no preconceived answer on this one.

The mathematics is not new. The monitoring application may be. If that sentence is wrong, this is the right room to kill it.

Now — every plate the engine produced

What follows is not slides about data. It is the data, run through the engine, scrolled through as a movie.

Nine countries. Twenty-six years. Six plates per country.

Master cover → 9 country sections · each section:

- cover slide
- Stage 1 Section plate — bearings, CLR, plan view
- System course plot — CLR trajectory PCA (the navigation chart)
- Helmsman family — Sigma sequence, flip count, stability
- ILR-Helmert Triplet plate — three orthogonal scatter panels
- CNQ dashboard — quaternion-view diagnostics

Sixty-six slides. Pause me anywhere. Every quantity hash-chained to the input CSV.

→ `CodaWork2026_PremierDataOutput_2026-05-13.pptx`

Q & A — with the manifold projector running

Live HTML projector. Interactive 3-D view of the energy-mix manifold. Stays on through the discussion.

WHILE YOU ASK QUESTIONS

- ORBIT — 3-D rotation of the manifold
- TRAILS — show the trajectory paths over time
- LABELS — year labels along each path
- GHOST — partial-transparency overlay across countries
- FRONT / SIDE / TOP — fixed projections

Run all · Run AUS / CHN / DEU / FRA / GBR / IND / JPN / USA / WLD individually.

Thank you. I'll take questions.

→ `codawork2026_projector.html` (open in browser; runs offline; no network required)

Repositories • reproduce every plate in five minutes

Hashes carry the receipts. Anyone with the raw EMBER CSV can verify any plate.

higgins-decomposition (H^s)

Deterministic engines • CNT v3.1.0 / CNQ v2.0.0
63 Investigation Catalog entries
H^s Change Control v1.0 • HUF-STD-001/002/003
Manuscript: papers/codawork2026/manuscript/

Higgins-Unity-Framework (HUF)

MC-4 framing • EITT canonical
HUF Governance Charter
KILL-001 • 19 named failure modes
Companion: case studies + cross-AI archive

[*github.com/PeterHiggins19/higgins-decomposition*](https://github.com/PeterHiggins19/higgins-decomposition)

[*github.com/PeterHiggins19/Higgins-Unity-Framework*](https://github.com/PeterHiggins19/Higgins-Unity-Framework)

Self-discipline note: the L2 → TV correction is on the record. We tell on ourselves when we mislabel.

The mathematics is not new. The monitoring application may be.

The talk is an ascent waypoint, not the summit.

AI Use Declaration

Per HUF Publication Standards (HUF-STD-001 v1.1) — conforming to scientific community conventions.

STANDARDS CONFORMANCE

ICMJE · COPE · Nature / Springer · Science / AAAS · WAME · EU AI Act (2024) · arXiv · ACM · IEEE

AI TOOLS USED

Claude (Anthropic) · ChatGPT (OpenAI) · Copilot (Microsoft) · Gemini (Google) · Grok (xAI)

Collectively: the HUF AI Collective.

TASKS PERFORMED BY AI

Drafting of supporting documents and slides · consistency editing · cross-checking claims across documents
Literature-search assistance for prior-art areas · adversarial review for overclaim and drift catching

AUTHOR RESPONSIBILITY

P. Higgins retains full responsibility for all scientific claims, data interpretation, methodological choices, and conclusions. All AI-generated content has been reviewed and verified. AI tools are NOT listed as authors.

H^s Framework • Engine Declaration

Standard Stamp — appended to every H^s-produced document • HUF-STD-001 compliant

From « CoDaWork 2026 — Final Talk — 2026-05-17 »

The Framework

Higgins Decomposition (H^s)
Deterministic compositional inference
Operates on the simplex in Aitchison geometry

Principle
CNT measures invariance.
CNQ names the algebra it lives in.

Open • Hash-chained • Reproducible
Apache-2.0 code + CC BY 4.0 docs
Same input → same output, always.

Validated across
11 domains • 101 reference datasets
44 orders of magnitude in scale
Three IEEE-floor confirmations:
drive failures, CMB photons,
Standard-Model neutrino oscillation.

Engines • Methods

CNT • Compositional Navigation Tensor
v3.1.0 (schema 3.1.0)
cnt.py (Python) + cnt.R (R port)
Closure → CLR → ILR-Helmert →
bearing tensor, angular velocity,
Higgins scale, depth tower,
IR class, attractor fit.

CNQ • Compositional Navigation Quaternion
v2.0.0 (schema cnq/2.0.0)
cnq.py (Python) + cnq.R (R port)
Quaternion view • CHSH joint-coherence
Twin-quaternion factoring at D=8

Plate suite • HUF-STD-002 link 4
Stage 0 Foundations (HUF-STD-003)
Stage 1 Section + ILR-Helmert Triplet
Stage 2 Helmsman + course + CoDa-PCA
Stage 3 Depth tower + attractor + κ^H s
Outputs: PDF / PNG / SVG

Find us • Contact

Repository
github.com/PeterHiggins19/
higgins-decomposition

Quick start (AI or researcher)
Load HS_FAST_REFRESH.json first
Read AI_AGENTS.md for the protocol
Reproduce with one command
per REPRODUCIBILITY_CHECKLIST.md

Sibling repositories
Higgins-Unity-Framework (HUF parent)
Rogue-Wave-Audio (DADC lineage)

Author • Lab
Peter Higgins
Rogue Wave Audio /
Binaural Test Lab
Markham, Ontario, Canada
RWA-001 (lab identity card)

Contact
PeterHiggins@RogueWaveAudio.com
Open an issue on GitHub
Help is available — free, no gatekeeping.

The instrument reads. The expert decides. The hashes carry the receipts. The vocabulary holds the line.